

Soil- and root properties in driving preferential flow of north subtropical forest stands: insights from machine learning

Yiyan Liu^{a,b,c}, Le You^{a,b,c}, Wenqi Zhang^{a,b,c}, Zhiying Tang^{a,b,c},
Di Wang^{a,b,c}, Yinghu Zhang^{a,b,c,*}

^a School of Forestry, Nanjing Forestry University, Nanjing 210037, PR China

^b Collaborative Innovation Center of Sustainable Forestry in Southern China of Jiangsu Province, Nanjing Forestry University, Nanjing 210037, PR China

^c Jiangsu Provincial Key Laboratory of Soil Erosion and Ecological Restoration, Nanjing Forestry University, Nanjing 210037, PR China

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ABSTRACT

As an important water infiltration property in the forest hydrological responses, the occurrence of preferential flow could be mainly explained by soil- and root properties. However, the mechanism of soil- and root properties influencing preferential flow at different forest stands remains unclear. In this study, dye-tracing experiments, machine learning algorithms, and shapley additive explanations approach were employed to analyze their relationships at the three forest stands (pine, bamboo, and oak). The results showed that preferential flow index (*PFI*) increased with increasing soil depth. *PFI* in the topsoils (0–20 cm) was in the following order: oak > pine > bamboo, while *PFI* in the subsoils (20–50 cm) was pine > oak > bamboo. The results also indicated that the Support Vector Machine (SVM) model demonstrated the excellent performance in explaining the complex relationships between soil properties, root properties, and preferential flow. The root volume showed the highest contribution to the predicted *PFI* at the pine forest stand, while soil capillary porosity and root biomass at the bamboo and oak forest stands, respectively. Soil bulk density with low values, also initial soil moisture, soil capillary porosity, soil non-capillary porosity, clay, and sand with high values could interact with root properties, thus influencing the changes in *PFI* jointly. The findings offer scientific basis for soil–water management strategies in forest ecosystems.

1. Introduction

Preferential flow refers to the fast transportation of solutes and energy through the soil vadose zones, bypassing most of the soil matrix (Nimmo, 2020a; Beven and Germann, 2013). In general, preferential flow mainly includes macropore flow, fingered flow, and funneled flow (Zhang et al., 2024; Nimmo, 2020b; Guo and Lin, 2017). As a common pattern of water movement, preferential flow could exert the inconsistent (positive or negative) influences on the eco-hydrological processes and other environmental issues. For instance, soil organic matter in preferential flow paths is younger, and the soil nitrogen cycling in preferential flow paths occurs faster than in the soil matrix (Bundt et al., 2001). Also, well-connected preferential flow paths could enhance soil permeability and improve soil physical–chemical properties (Zhao et al., 2024; Zhang et al., 2018; Bogner et al., 2012). More ecosystem services, such as flood regulation, erosion controlling, and aquifer recharge, could also benefit from preferential flow paths (Guo and Lin, 2017). Besides,

the negative effects of preferential flow could not be neglected. Preferential flow could destabilize slopes, trigger landslides, accelerate gully erosion by eroding soil pipes (Xu et al., 2024; Zhang et al., 2021b; Zhang et al., 2018), and cause groundwater pollution by rapidly transporting pollutants to the groundwater table (Ge et al., 2025; Allaire et al., 2016; Villamizar and Brown, 2016). Preferential flow might facilitate the transport of soil nutrients to the subsoils, which could avoid their adsorption to the surrounding soil matrix (Malhotra et al., 2024; Makowski et al., 2020), resulting in the nutrients loss and even soil degradation (Li et al., 2024). Therefore, accurate understanding the driving mechanisms of preferential flow could better recognize their positive or negative influences in pedological perspectives.

Promising studies have shown that the occurrence of preferential flow is influenced by various driving factors, in particular plant root systems, rock fragments, and soil properties (Guo and Lin, 2017). For instance, fine root systems significantly impact the occurrence of preferential flow compared to coarse root systems (Zhang et al., 2015). The

* Corresponding author at: School of Forestry, Nanjing Forestry University, Nanjing 210037, PR China.

E-mail address: zhang9c415@njfu.edu.cn (Y. Zhang).

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positive impacts of fine root systems could be attributed to the root preferential channels during root decomposition (Guo et al., 2019). In one respect, root systems rearrange the distribution of soil particles by fully contacting, dividing, and squeezing the soils, thereby increasing soil porosity and influencing preferential flow (Cai et al., 2024; Shi et al., 2021; Fischer et al., 2014). In another respect, root enlargement could be impeded with increasing soil bulk density, such reducing the occurrence of macropores between root systems and the surrounding soils (de Moraes et al., 2020). Rock fragments could promote the occurrence of preferential flow when the total volume fraction is < 30 %, while when the total volume fraction exceeds 30 %, rock fragments could reduce preferential flow or have no impacts (Hou et al., 2022). Preferential flow might also result from spatial variations in soil properties with increasing soil depth (Zhang et al., 2024). The areas with high sand content and low initial soil moisture might facilitate the occurrence of preferential flow, whereas areas with low sand content but high initial soil moisture are more likely to promote the occurrence of preferential flow (Kang et al. 2022). Ding et al. (2023) reported that sand content was significantly positively correlated with preferential flow, while silt content was significantly negatively correlated with preferential flow, and clay content showed no impacts on preferential flow. Nevertheless, Grant et al. (2019a) indicated that the preferential flow intensity increased with increasing clay content. With respect to soil bulk density, Koestel et al. (2012) and Mossadeghi-Björklund et al. (2015) found a negative correlation between soil bulk density and preferential flow, while Katuwal et al. (2015) and Paradelo et al. (2016) concluded that the strength of preferential transport was positively correlated with soil bulk density. Studies concluded that the role of soil porosity in the occurrence of preferential flow might be contradictory. Increased soil total porosity and macroporosity could enhance infiltration capacity and preferential flow (Qiu et al., 2022). A high soil porosity indicates a large interfacial area between pores and the soil matrix, allowing more water to infiltrate into the matrix, which also inhibits the occurrence of preferential flow (Kang et al., 2022; Jarvis, 2007). However, the occurrence of preferential flow is influenced by many small-scale soil- and root properties heterogeneities, whose roles are still uncertain and vary by site (Zhang et al., 2024; Kang et al., 2022). In short, due to vertical soil- and root properties heterogeneity and diverse external conditions at different sites, their influences driving the changes in the occurrence of preferential flow are still unclear.

The complex, non-linear interaction between natural environmental factors and preferential flow poses a challenge for traditional analytical methods (Kang et al., 2023; Liu and Lin, 2015). Recent studies have addressed the limitation by evaluating the performance of the machine learning algorithms, which indicates that the potential of machine learning algorithms to capture complex interaction is more reliably than traditional approaches (Mazighi et al., 2025). In fact, there are lots of promising machine learning algorithms applied to explore the relationships between different variables and preferential flow. Six machine learning algorithms among them, such as XGBoost, Random Forest (RF), CatBoost, Gradient Boosting Decision Tree (GBDT), Light Gradient Boosting Machine (LGBM), and SVM, were rationally selected. For example, Li et al. (2025) used the XGBoost, RF, and SVM to identify preferential flow paths and concluded the accuracy for the models. Mazighi et al. (2025) applied CatBoost, GBDT, and LGBM to predict the preferential infiltration rate. Kocian and Mohanty (2024) employed the RF to predict preferential flow events and showed that the occurrence of preferential flow events had seasonal dependence, with the highest occurrence rate in spring. However, regression problems often involve highly complex internal and external factors, and different machine learning algorithms might vary in predictive performance (Huang et al., 2019). Shapley additive explanations (SHAP) based on a game theory-based approach could determine the positive or negative contribution of each input variable to a prediction (Wu and Zhou, 2022; Mangalathu et al., 2020). Using SHAP can better help us understand the model's prediction results and the contributions of each driving factor.

Therefore, identifying the optimal model from different machine learning algorithms and fully understanding the decision-making process of algorithms to comprehensively reveal the complex relationship between soil- and root properties and preferential flow becomes more urgent.

Dye tracers have been widely used in visualizing preferential flow paths in the field (Weiler and Flühler, 2003). Brilliant Blue FCF with high visibility and low toxicity is the most popular dye in the studies of preferential flow (Wang et al., 2024; Zhang et al., 2022). In this study, we selected three typical subtropical forest stands—pine (*Pinus taeda*), oak (*Quercus acutissima*), and bamboo (*Phyllostachys edulis*)—located in the eastern China as the research objects to certify the effects of soil- and root properties on the occurrence of preferential flow utilizing dye-tracing experiments. We hypothesized that the influences of soil- and root properties on preferential flow were complex and their driving mechanism was various at different forest stands. This study aimed to (1) clarify the variation of soil- and root properties in the soil depth and the differences among the three forest stands; (2) visualize the differences in preferential flow among the three forest stands; (3) quantify the influence of soil- and root properties on preferential flow and explore their driving mechanisms. This study enhances our knowledge of soil hydrological processes in forest ecosystems and provides a scientific support for reasonable forestry management. The flowchart of this study was shown in Fig. 1.

2. Materials and methods

2.1. Study area

The study area is located (Fig. 2) in Xiashu Forest Farm, which is affiliated to Xiashu Town, Zhenjiang City, Jiangsu Province (119°14' E, 32° 08' N). The highest peak in Xiashu Forest Farm reaches 377.8 m above sea level, with a relative height difference of 302.8 m. It has a north subtropical monsoon climate, featuring an annual average temperature of 15.2, an annual average precipitation of 1055.6 mm, and significant interannual variation in precipitation. According to the International Soil Texture Classification System Triangle Coordinate Map, the soil classification in the study area is the yellow-brown loam and the soil depth is within the soil layers from 0–50 cm (Chen et al., 2025; Tang et al., 2024). The types of soil stoniness mainly include sedimentary rocks, magmatic rocks, and metamorphic rocks. The area has an annual average of 2157 h of sunshine and a sunshine percentage of 4 %. The forest coverage rate in the forest farm exceeds 90 %. The main tree species include the sawtooth oak (*Quercus acutissima* Carruth.), bamboo (*Phyllostachys edulis* (Carrière) J. Houz.), Masson's pine (*Pinus massoniana* Lamb.), Chinese cork oak (*Quercus variabilis* Blume.), Chinese sweetgum (*Liquidambar formosana* Hance.), Chinese hackberry (*Celtis sinensis* Pers.), Zelkova (*Zelkova serrata* (Thunb.) Makino), among others. The field investigation showed that the pine and oak forests are deep rooted species, while the dense fibrous roots in bamboo forests are mainly horizontally distributed in the topsoils.

2.2. Soil sampling and laboratory analysis

The experiments were conducted in the autumn of 2021. Three sites were established for pine, bamboo, and oak forest stands. The spatial distance between the pine, bamboo, and oak forest stands was less than 1 km to reduce soil heterogeneity. The more information about sites was shown in Table 1. Three replicate plots of 20 m × 20 m were installed at each site. The five-point sampling method was used at each sampling point. 135 soil cores (height: 10 cm, diameter: 7.2 cm, volume: 400 cm³) were taken at three forest stands to collect root samples from soil depths of 0–10 cm, 10–20 cm, 20–30 cm, 30–40 cm, and 40–50 cm. Then, the soil cores should be placed into water to separate the soil particles from the root systems. The root systems were placed on a sieve under running water to rinse off the soil particles remaining on the root surfaces. After

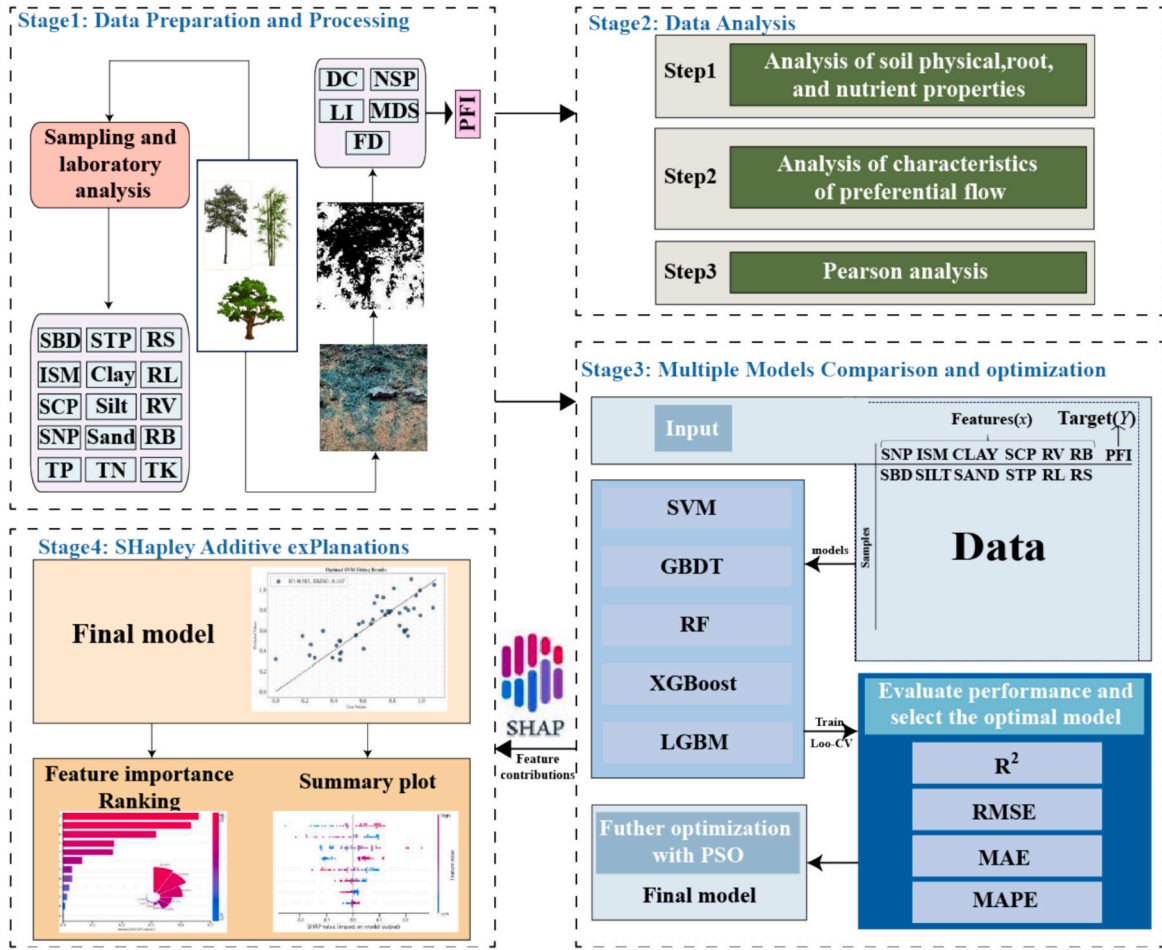


Fig. 1. The flowchart of this study.

the root systems were air-dried for one week, they were scanned and analyzed using the WinRHIZO Pro root analysis system to get the root length (RL), surface area (RS), and volume (RV). Those root systems were dried and weighed to get the root biomass (RB) (Wang and Zhang, 2009). Meanwhile, soil samples were also extracted from the five soil depths to measure soil physical properties and soil nutrients content. A total number of 2025 soil samples were collected (3 forest stands $\times$ 3 sites $\times$ 3 sampling plots $\times$ 5 points $\times$ 3 replicates $\times$ 5 depths). The methods for measuring soil physical- and soil nutrients were shown in Table 2. All analytical statistics were derived from $n = 45$ independent samples.

2.3. Dye-tracing method

2.3.1. Field dye-tracing experiments

In this study, Bright Blue FCF was selected as the dye tracer, and field dye-tracing experiments were conducted at each site. A total number of nine $70 \text{ cm} \times 70 \text{ cm}$ plots were established at each forest stand. Surface litter and vegetation were removed from the plot. A 50 L dye solution (4 g/L Bright Blue FCF in deionized water) was poured into a simulated rainfall apparatus and evenly sprayed homogeneously onto the soil surface of the plot. After spraying, the plot surface was covered with plastic film. The soil profile was excavated to the soil depth of 50 cm because of the bedrock layers below the soil depth. Then, the dye-stained soil profiles were photographed using a digital camera twenty-four hours later, such enabling accurate visualization of preferential flow.

2.3.2. Image processing and preferential flow index calculation

In this study, a $50 \text{ cm} \times 50 \text{ cm}$ dye-stained area was analyzed using Adobe Photoshop 2020. The analytical image was then pixelated to a resolution of 500×500 pixels, with the dye-stained areas replaced by black (0) and the non-dye-stained areas replaced by white (255) by binary processing. Noise reduction was applied to the image using the Process-Filter-Morphological function in Image-Pro Plus. The dye coverage (DC), length index (LI), fractal dimension (FD), the number of dye-stained patches (NSP), and the mean diameter of dye-stained patches (MDS) are important parameters of preferential flow. By analyzing these parameters, we could evaluate preferential flow intensity.

The DC is the ratio of the dye-stained areas to the total areas of the soil profiles. The larger the ratio of the dye-stained areas, the lower preferential flow intensity.

$$DC = \frac{D}{D + ND} \quad (1)$$

where D is the dye-stained areas; ND is the non-dye-stained areas.

The LI is the sum of the absolute differences between the DC_{i+1} (%) and DC_i (%) of two consecutive layers in a vertical soil profile.

$$LI = \sum_{i=1}^n |DC_{(i+1)} - DC_i| \quad (2)$$

n is the total number of soil layers in a vertical soil profile, DC_{i+1} is the ratio of the dye-stained areas of the $i + 1$ th layer in a vertical soil profile, and DC_i is the ratio of the dye-stained areas of the i th layer in a vertical soil profile.

The FD provides a quantitative analysis of structural complexity for

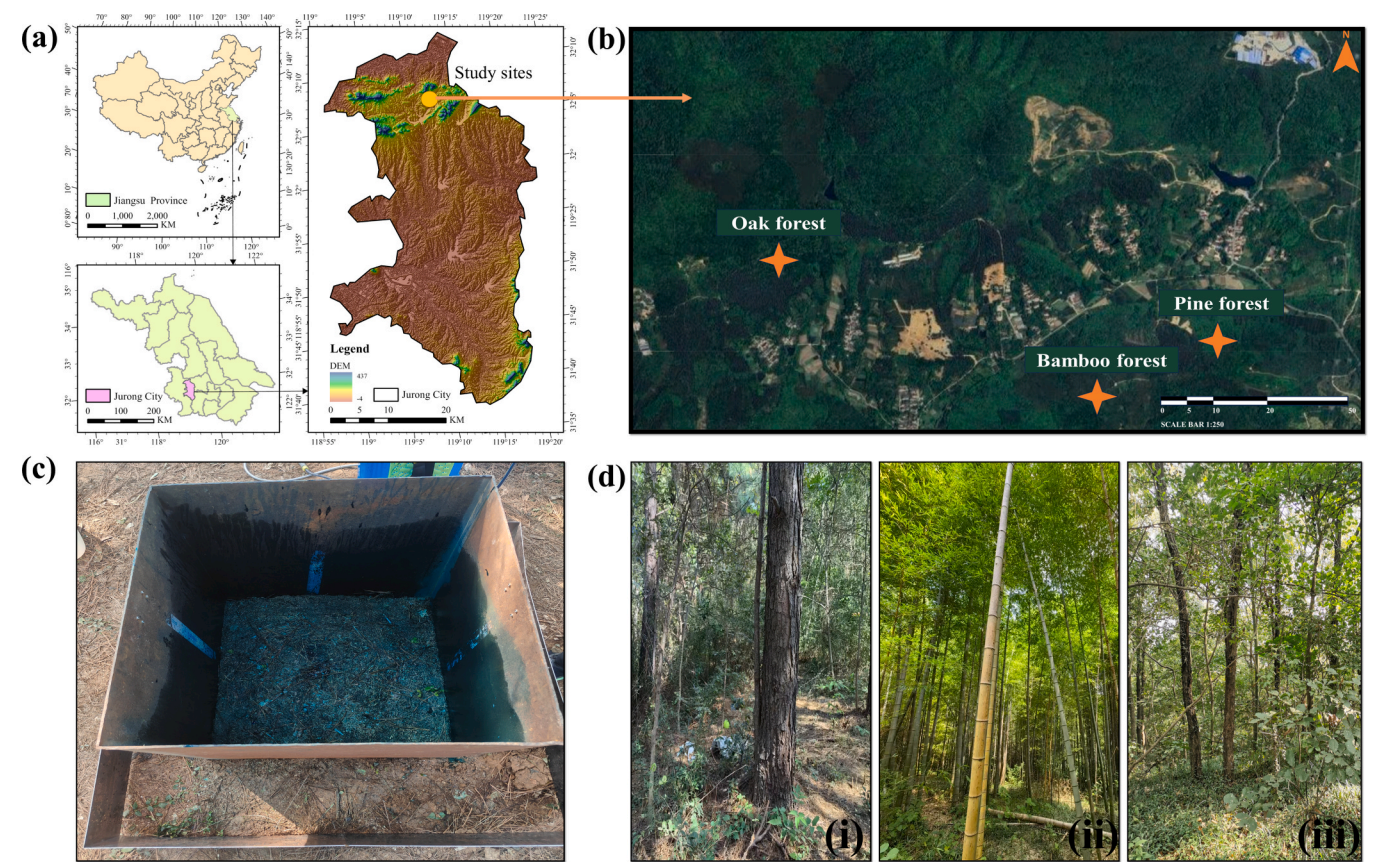


Fig. 2. (a) The location of the study area. (b) The Location of experimental plots. (c) Dye-tracer experiments. (d) Different vegetation types. (i): pine forest stand; (ii): bamboo forest stand; (iii): oak forest stand.

Table 1
Basic information of sampling forest stands (Chen et al., 2025).

Forest stands	Major vegetation	Age/ year	Longitude and latitude	Average diameter breast height/cm	Average tree height/m	Coverage /%	Forest density/ hm ²	Soil type	Vegetation composition
Pine	<i>Pinus taeda</i> L.	47	119°13'19"E 32°7'15"N	29.47	18.37	78	0.46	Yellow-brown loam	<i>Camellia sinensis</i> (L.) O. Kuntze, <i>Boehmeria nivea</i> (L.) Gaudich., <i>Lygodium japonicum</i> (Thunb.) Sw., <i>Rubus lambertianus</i> Ser., <i>Opilsenus undulatifolius</i> (Arduino) Roem. & Schult., etc.
Bamboo	<i>Phyllostachys edulis</i> (Carriere) J. Houzeau	42	119°13'9"E 32°7'8"N	7.11	11.57	92	0.12	Yellow-brown loam	<i>Rubus corchorifolius</i> L. f., <i>Trachelospermum jasminoides</i> (Lindl.) Lem., <i>Smilax china</i> L., <i>Lygodium japonicum</i> (Thunb.) Sw., etc.
Oak	<i>Quercus acutissima</i> Carruth	52	119°12'12"E 32°7'40"N	38.85	21.40	84	0.23	Yellow-brown loam	<i>Phyllanthus glaucus</i> Wall. ex Müll. Arg., <i>Trachelospermum jasminoides</i> (Lindl.) Lem., <i>Smilax china</i> L., <i>Lindera glauca</i> (Sieb. et Zucc.) Blume., <i>Ampelopsis brevipedunculata</i> (Maxim.) Trautv., etc.

irregular or self-similar patterns.

$$FD = \lim_{\varepsilon \rightarrow 0} [\log N(\varepsilon) / \log (1/\varepsilon)]$$

(3)

where ε is the average width of each dye-stained area; $N(\varepsilon)$ is the fractal number covered by the dye-stained areas.

In this study, the *NSP* and *MDS* could be calculated through the Measure-Count/Size function in Image Pro Plus. The dye-stained patches represent the preferential flow paths (Zhang et al., 2024). Large number of dye-stained patches could lead to higher preferential flow intensity. Based on the statistical data, the threshold of mean diameter of dye-stained patches was 20 mm. Preferential flow dominated the soil water flow behavior when the *MDS* was < 20 mm, while

soil matrix flow dominated when the *MDS* was > 20 mm. In further, when the *MDS* was < 20 mm, the larger the *MDS*, the higher preferential flow intensity.

2.3.3. Construction of preferential flow index

(1) Dimensionless treatment of evaluation parameters

Preferential flow parameters (*DC*, *LI*, *FD*, *NSP*, and *MDS*) were normalized to be dimensionless. The calculation formula is as follows:

$$Z_{ij} = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}} \text{ (positive correlation parameter)}$$

(4)

Table 2

Methods of determining soil physical properties and nutrients.

Soil indicators	Analytical methods	Reference authors
Soil bulk density (<i>SBD</i>), Initial soil moisture (<i>ISM</i>), Soil capillary porosity (<i>SCP</i>), Soil non-capillary porosity (<i>SNP</i>), Soil total porosity (<i>STP</i>)	Ring cutter method	Wang et al. (2024)
Soil particle size distribution (i.e., sand, silt, and clay)	dry sieving and sedimentation analysis	Schutter and Dick (2002)
Soil total phosphorus (<i>TP</i>)	Continuous flow analytical system (SKALAR SAN++, The Netherlands)	Miranda et al. (2001)
Soil total nitrogen (<i>TN</i>)	Kjeldahl digestion distillation method	Bremner et al. (1960)
Soil total potassium (<i>TK</i>)	Flame photometry method	Banerjee and Prasad (2020)

$$Z_{ij} = \frac{X_{\max} - X_i}{X_{\max} - X_{\min}} \quad (\text{negative correlation parameter}) \quad (5)$$

Here, Z_{ij} represents the dimensionless standardized value of the i -th parameter for the j -th observation, X_i represents the actual value of the i -th parameter, $X_{\max}$ and $X_{\min}$ are the actual maximum and minimum values of the parameter, respectively.

(2) Construction of preferential flow index (*PFI*)

The mean value of each parameter $E(G_j)$ reflects a measure of central tendency of parameter. The calculation formula is as follows:

$$E(G_j) = \frac{1}{n} \sum_{i=1}^n Z_{ij} \quad (6)$$

The mean-squared deviation $\sigma(G_j)$ measures the degree of dispersion of parameter. The larger the $\sigma(G_j)$, the stronger a distinction between heterogeneity in parameters. The $\sigma(G_j)$ was calculated as follows:

$$\sigma(G_j) = \sqrt{\frac{1}{n} \sum_{i=1}^n (Z_{ij} - E(G_j))^2} \quad (7)$$

The weighting coefficient $W(G_j)$ shows the relative contribution of each parameter to the changes in the *PFI*. The larger the $W(G_j)$, the higher contribution of the parameter.

$$W(G_j) = \frac{\sigma(G_j)}{\sum_{i=1}^n \sigma(G_j)} \quad (8)$$

The *PFI* is assessed for stronger or lower preferential flow intensity. The higher *PFI*, the stronger preferential flow intensity. The *PFI* is calculated using the following formula:

$$PFI = \sum_{j=1}^m Z_{ij} W(G_j) \quad (9)$$

2.4. Comparison of models and SHapley additive exPlanations

2.4.1. Model training, evaluation and optimization

In this study, we selected six advanced machine learning models (i.e., the Support Vector Machine (*SVM*), Random Forest (*RF*), Gradient Boosting Decision Tree (*GBDT*), *CatBoost*, LightGBM (*LGBM*), and *XGBoost*) to conduct the impacts of soil- and root properties on preferential flow. The datasets were independent samples from soil- and root sampling, and laboratory analysis. The input variables for each model include soil- and root properties. These input variables together form a comprehensive set of features for predicting the output *PFI*. To ensure the accuracy and reliability of model evaluation, we used the leave-one-out cross-validation (*LOO-CV*) method to validate the models (Cawley and Talbot, 2003). The method uses each sample as the validation set in turn, and the remaining samples as the training set, thereby maximizing

the use of data and ensuring the unbiasedness of model evaluation (Cawley and Talbot, 2003). The evaluation indicators for model performance include the coefficient of determination (R^2), root mean square error (*RMSE*), mean absolute error (*MAE*), and mean absolute percentage error (*MAPE*). The higher R^2 , the lower *MAE* and *RMSE*, the better the model's performance. The range of *MAPE* values is $[0, +\infty)$. *MAPE* values less than 100 indicate that the model performance is reliable (Chicco et al., 2021).

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (10)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (11)$$

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n} \quad (12)$$

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{y_i} \quad (13)$$

where y_i is the actual values, $\hat{y}_i$ is the predicted values, and n is the number of samples.

After the initial evaluation of the model, we selected the best-performing model and further optimized its parameters using the Particle Swarm Optimization (*PSO*) algorithm. *PSO* is a swarm intelligence-based optimization technique inspired by the collective behavior of bird flocks and fish schools (Wang et al., 2017). It iteratively solves problems by simulating the cooperation and competition among individuals, thereby finding the optimal solution (Song and Gu, 2004). Ultimately, the model optimized by *PSO* will show a significant improvement in performance on the test set, comparing the model before optimization.

2.4.2. Shapley additive explanations (*SHAP*)

SHAP values could be used to quantify each feature's contribution to model predictions using Shapley value theory, thereby improving model interpretability. In this study, the explanation model $g(x')$ with simplified input x' for an original model $f(x)$ is as follows (Mangalathu et al., 2020):

$$f(x) = g(x') = \phi_0 + \sum_{i=1}^m \phi_i x'_i \quad (14)$$

where the m represents the numbers of input parameters, and x represents the input parameters ($x = x_1, x_2, \dots, x_p$). There is a relationship between x and x' through a mapping function, $x = h_x(x')$; ϕ_0 represents the constant value when all input parameters are missing.

$$\phi_i(f, x) = \sum_{z' \subseteq x'} \frac{|z'|!(M - |z'| - 1)!}{M!} [f_x(z') - f_x(z'/i)] \quad (15)$$

where $|z'|$ is the number of non-zero entries in z' , $f_x(z') = f(h_x(z')) = E[f(z)|z_S]$, and S is the *SHAP* values (Mangalathu et al., 2020; Lundberg and Lee, 2017).

2.5. Data analysis

We conducted a normality test for all the data, and performed one-way ANOVA to analyze the differences of soil- and root properties and *PFI* in different soil layers of the three forest stands with SPSS 26.0. All Significant differences were tested at $P < 0.05$. Pearson analysis was used to analyze the correlation between preferential flow and soil- and root properties, which performed by the Correlation Plot plugin from origin 2021. In this study, six machine learning models and *SHAP* were conducted through python 3.13.0 to reveal the driving mechanism. Various packages including "scikit-learn" "xgboost" "lightgbm" "catboost" "pyswarms" "shap" "matplotlib" and "seaborn" were considered. We adopted the *LOO-CV* method for model validation. To evaluate the

model's performance, we employed several key indicators such as the R^2 , $RMSE$, MAE , and $MAPE$. Furthermore, the PSO was utilized to optimize the best model. In final, $SHAP$ was implemented to accurately quantify each variable's contribution to the prediction results of the model.

3. Results

3.1. Quantification of soil- and root properties at the three forest stands

3.1.1. Characteristics of soil physical properties

Values of SBD at the time collection ranged from 0.985 to 1.485 g/cm³, while values of ISM ranged from 10.575 % to 30.039 % (Fig. 3). Average values of SBD were higher in the lower soil layers (1.355 g/cm³ at 10–50 cm) compared to the upper soil layers (1.052 g/cm³ at 0–10 cm), while average values of ISM and soil porosity were higher in the upper soil layers (ISM : 23.49 %, STP : 52.36 %, SCP : 43.31 %, SNP : 9.05 % at 0–10 cm and ISM : 19.95 %, STP : 44.08 %, SCP : 39.57 %, SNP : 4.51 % at 10–50 cm). SBD was not significantly different among the three forest stands, while ISM at the oak forest stand was significantly different from the bamboo and pine forest stands ($P < 0.01$). Soil porosity (STP , SCP , and SNP) decreased with increasing the depth of the soil. SCP at the oak forest stand was different from the pine and bamboo forest stand significantly ($P < 0.05$), while there were no significant differences among the three forest stands for SNP and STP ($P > 0.05$). Textural analysis of soil samples showed that the most abundant soil fraction was silt. There were no significant differences with respect to the content of sand and silt among the three forest stands in all soil layers ($P > 0.05$), while the clay content at the pine forest stand at the soil depth of 30–40 cm and 0–10 cm was significantly different from that at the bamboo forest stand ($P < 0.01$). There were no significant differences with respect to clay content between the bamboo and oak forest stand in all soil layers ($P > 0.05$).

3.1.2. Characteristics of soil nutrient properties

The values of TN and TP in topsoils (TN : 1.429 g/kg and TP : 215.61 mg/kg at soil depth of 0–10 cm) were larger than those in the lower soil layers (TN : 0.590 g/kg and TP : 164.58 mg/kg at the soil depth of 10–50 cm). The difference between the pine and bamboo forest stands was significant for TN at the soil depth of 10–20 cm and 40–50 cm ($P < 0.05$) (Fig. 3). TN at the pine forest stand was significantly different from the other forest stands at the soil depth of 30–40 cm ($P < 0.05$). TP in topsoils at the pine forest stand was significantly different from that at the bamboo and oak forest stands ($P < 0.05$). The values of TK increased with increasing soil depth at the three forest stands (Fig. 3). The differences among the three forest stands for TK were significant in topsoils (0–10 cm) ($P < 0.05$). TK at the oak forest stand was significantly different from that at the pine and bamboo forest stands at the soil depth of 10–30 cm and 40–50 cm ($P < 0.01$). The difference between the pine forest stand and the other forest stands for TK was significant at the soil depth of 30–40 cm ($P < 0.001$). TN and TK in topsoils (0–10 cm) were significantly different from those in lower soil layers (10–50 cm) at the three forest stands ($P < 0.01$), while TP in topsoils was not significantly different from that in lower soil layers ($P > 0.05$).

3.1.3. Characteristics of root properties

The values of root properties in the study area were higher in the upper soil layers (RL : 272.58 cm/100 cm³, RS : 41.13 cm²/100 cm³, RV : 0.516 cm³/100 cm³, RB : 0.418 g/100 cm³ at soil depth of 0–20 cm) compared to the lower soil layers (RL : 90.45 cm/100 cm³, RS : 13.19 cm²/100 cm³, RV : 0.174 cm³/100 cm³, RB : 0.145 g/100 cm³ at 20–50 cm). The values of root properties at the three forest stands decreased with increasing soil depth (Fig. 3). RL at the bamboo forest stand was significantly different from the pine and oak forest stands at the soil depth of 0–20 cm and 40–50 cm ($P < 0.01$). RS , RV , RB at the bamboo forest stand in the upper soil layers were significantly different from

those at the pine and oak forest stands ($P < 0.01$). However, there were no significant differences among the three forest stands in other soil layers ($P > 0.05$). RL , RS , RV , and RB in the upper soil layers (0–20 cm) were significantly different from those in lower soil layers (20–50 cm) at the three forest stands ($P < 0.05$).

3.2. Characterization of preferential flow index (PFI) at the three forest stands

The PFI was calculated by different measured preferential flow parameters in the study area (Table 3). The DC and the NSP showed the higher weighting coefficient compared to other preferential flow parameters, which indicated that the two preferential flow parameters could better imply the changes in the PFI (Table 4). The PFI as a whole increased with increasing soil depth at the three forest stands. PFI in topsoil (0–20 cm) was in the following order: oak > pine > bamboo, while PFI in subsoil (20–50 cm) was pine > oak > bamboo (Fig. 4). Values of the PFI at the pine forest stand ranged from 0.31 to 0.988, 0.14 to 0.81 at the bamboo forest stand, and values of the PFI ranged from 0.63 to 0.83 at the oak forest stand. Average values of the PFI were higher in the lower soil layers (0.91 at 20–50 cm) compared to the upper soil layers (0.36 at 0–20 cm) at the pine forest stand. Average values of the PFI were higher in the lower soil layers (0.78 at 20–50 cm) compared to the upper soil layers (0.30 at 0–20 cm) at the bamboo forest stand. Average values of the PFI were higher in the lower soil layers (0.78 at 20–50 cm) compared to the upper soil layers (0.64 at 0–20 cm) at the oak forest stand. At the bamboo and pine forest stands, the PFI in the upper soil layers (0–20 cm) was significantly different from that in the lower soil layers (20–50 cm) ($P < 0.01$), while there was no significant difference among different soil layers at the oak forest stand ($P > 0.05$). There were no significant differences for the PFI among the three forest stands in different soil layers ($P > 0.05$).

3.3. Relationships between soil properties, root properties, and preferential flow

The Pearson analysis of effects of soil- and root properties on PFI was revealed in detail (Figs. 5a, b, c). PFI was significantly positively correlated with SBD ($P < 0.001$), while it was significantly negatively correlated with ISM ($P < 0.05$), SCP ($P < 0.01$), SNP ($P < 0.05$), STP ($P < 0.001$), and root properties (RL , RS , RV , and RB) ($P < 0.001$) at the pine forest stand. At the bamboo forest stand, PFI was significantly positively correlated with SBD ($P < 0.001$), while it was significantly negatively correlated with SCP ($P < 0.01$), STP ($P < 0.01$), and root properties (RL , RS , RV , and RB) ($P < 0.001$). PFI at the oak forest stand was significantly negatively correlated with SCP ($P < 0.05$), RL ($P < 0.05$), RS ($P < 0.01$), RV ($P < 0.05$), RB ($P < 0.001$), while other soil properties showed no significant correlations ($P > 0.05$). PFI was positively correlated with TK ($P < 0.001$), while it was negatively correlated with TN and TP at the three forest stands.

3.4. Comparison of different machine learning models and SHAP explanation

The results showed that the SVM model performs the best in terms of the correlation between the actual and predicted values at the three forest stands (Figs. 6, 7). The SVM model was further optimized by the PSO algorithm. The SVM model after optimization showed a higher correlation coefficient ($R^2 = 0.632$, $R^2 = 0.638$, $R^2 = 0.733$ at the pine/bamboo/oak forest stands, respectively) and smaller error parameters ($RMSE = 0.173$, $RMSE = 0.178$, $RMSE = 0.077$ at the pine/bamboo/oak forest stands, respectively) compared with the SVM model before optimization (Fig. 8). The results indicated that RV has the highest SHAP value at the pine forest stand, while SCP and RB have the highest SHAP values at the bamboo- and oak forest stand (Fig. 9). The results revealed that PFI at the pine forest stand might be partly explained by a

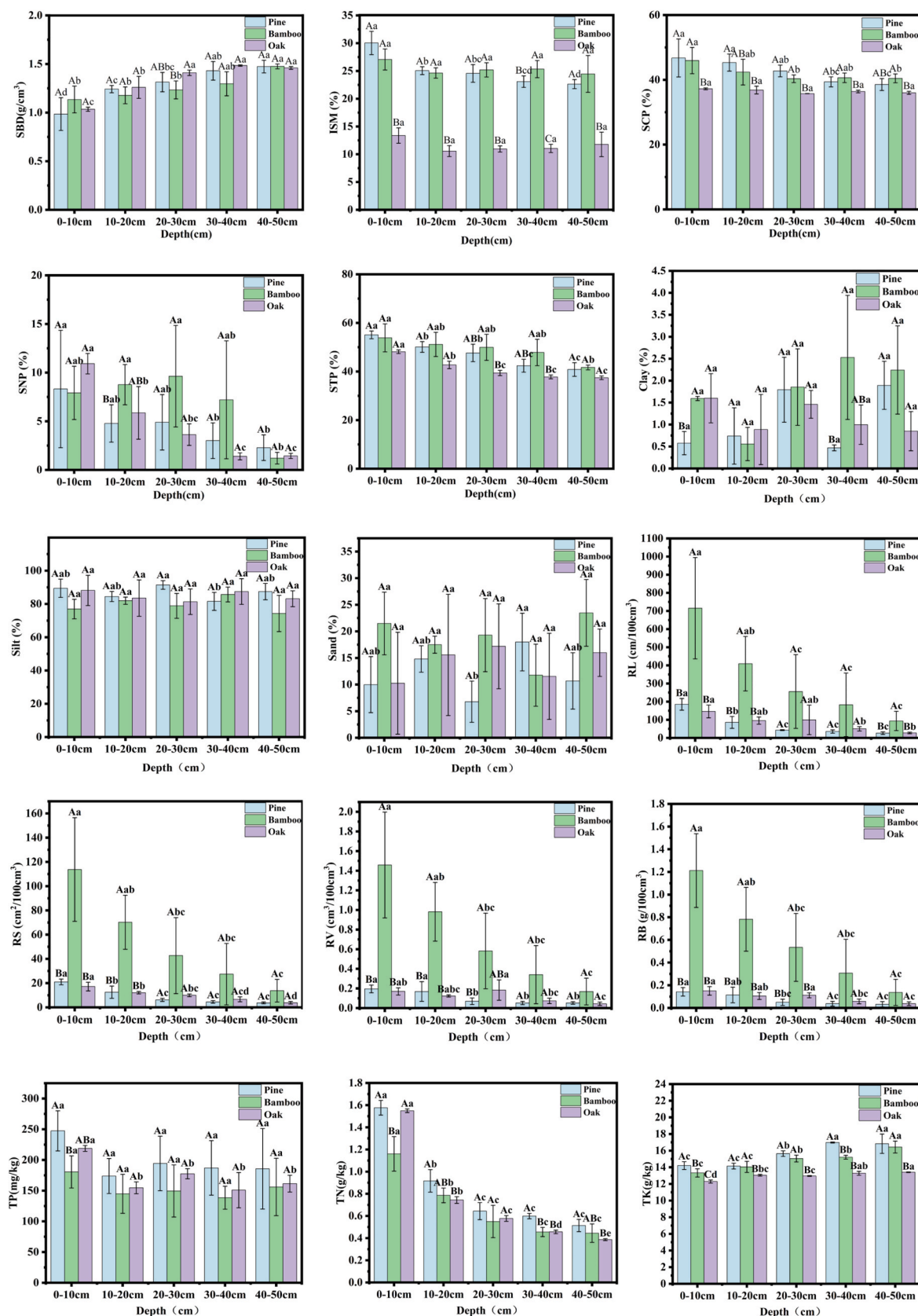


Fig. 3. Changes of soil physical properties, root properties, and soil nutrients in different soil layers at different forest stands. SBD: Soil bulk density; ISM: Initial soil moisture; SCP: Soil capillary porosity; SNP: Soil non-capillary porosity; STP: Soil total porosity; RL: Root length; RS: Root surface area; RV: Root volume; RB: Root biomass; TP: Soil total phosphorus; TN: Soil total nitrogen; TK: Soil total potassium. Capital letters indicate significant differences at different forest stands in the same soil layer. Lowercase letters indicate significant differences in different soil layers at the same forest stand.

Table 3

Measured preferential flow parameters at the three forest stands.

Forest stands	Soil depth (cm)	DC (%)	LI (%)	FD	NSP	MDS (mm)
Pine	0–10	91.51 ± 6.25Aa	78.34 ± 29.60Bcd	1.35 ± 0.16Bc	31.54 ± 23.89Bc	1.42 ± 0.73Bc
	10–20	90.51 ± 4.37Aa	90.51 ± 26.18Bc	1.39 ± 0.14Ac	52.73 ± 24.80Bbc	1.96 ± 0.69Ac
	20–30	55.98 ± 10.86Ab	150.23 ± 21.99Aa	1.72 ± 0.04Aa	231.50 ± 83.15Aa	3.06 ± 0.98Ab
	30–40	28.02 ± 16.88Ac	122.87 ± 29.46Ab	1.60 ± 0.11Ab	232.73 ± 74.78ABa	3.58 ± 0.91Aab
	40–50	7.12 ± 6.21Ad	61.62 ± 30.84Cd	1.44 ± 0.12Ac	80.64 ± 51.76Ab	4.00 ± 1.71Aa
Bamboo	0–10	95.19 ± 4.84Aa	36.88 ± 24.67Cb	1.17 ± 0.47Bb	6.99 ± 4.61Bc	1.23 ± 0.61Cc
	10–20	80.96 ± 11.13Bab	80.14 ± 35.24Bab	1.54 ± 0.16Ba	53.11 ± 86.33Bbc	2.08 ± 0.65Ab
	20–30	62.65 ± 15.23Ab	109.69 ± 38.57Ba	1.67 ± 0.06Ba	126.67 ± 119.72Bab	2.95 ± 0.89Aa
	30–40	35.74 ± 16.70Ac	99.13 ± 33.49Ba	1.66 ± 0.14Aa	175.22 ± 156.89Ba	3.07 ± 0.72Aa
	40–50	19.69 ± 15.54Bc	82.74 ± 37.43Bab	1.55 ± 0.21Aa	152.72 ± 123.04Bb	3.34 ± 1.03Ba
Oak	0–10	70.87 ± 14.13Ba	148.41 ± 40.74Aa	1.60 ± 0.11Aa	218.33 ± 160.86Aa	2.51 ± 0.52Aa
	10–20	64.84 ± 22.16Ca	127.77 ± 14.47Ab	1.54 ± 0.10Bab	216.28 ± 168.27Aa	2.63 ± 0.70Aa
	20–30	61.51 ± 19.75Aa	138.11 ± 28.20Aab	1.58 ± 0.11Cab	238.83 ± 121.23Aa	2.72 ± 0.58Aa
	30–40	46.74 ± 17.66Ab	133.56 ± 22.76Aab	1.59 ± 0.11Aab	286.94 ± 113.15Aa	2.95 ± 0.67Aa
	40–50	33.39 ± 19.18Cc	119.68 ± 20.99Ab	1.51 ± 0.12Ab	251.61 ± 104.54Aa	2.90 ± 0.62Ba

Capital letters indicate significant differences in preferential flow parameters at different forest stands in the same soil layer; lowercase letters denote significant differences in preferential flow parameters in different soil layers at the same forest stand.

Table 4

Standardized mean, mean square deviation and weight of preferential flow parameters.

Preferential flow parameters	E(G _j)	σ(G _j)	W(G _j)
DC	0.432	0.302	0.363
LI	0.459	0.225	0.271
FD	0.731	0.177	0.213
NSP	0.402	0.296	0.355
MDS	0.359	0.202	0.243

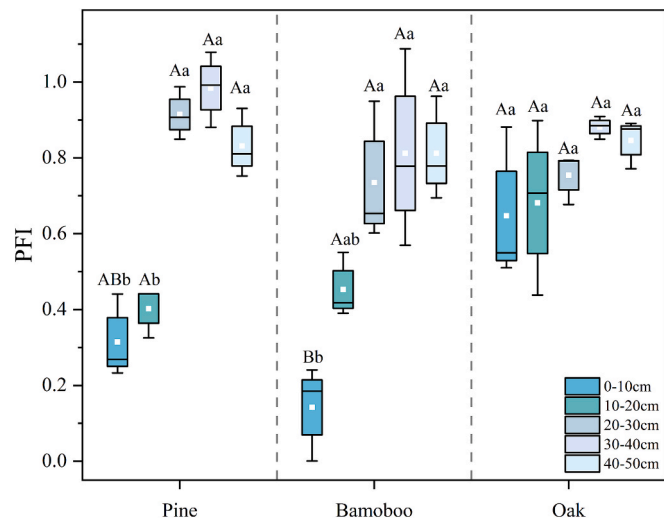


Fig. 4. Changes of PFI in different soil layers at different forest stands. Capital letters indicate significant differences in PFI at different forest stands in the same soil layers. Lowercase letters indicate significant differences in PFI in different soil layers at the same forest stand.

combination of RV and SBD (Fig. 9). At the bamboo forest stand, PFI might be explained by a combination of RL and SCP. PFI at the oak forest stand could be partly explained by a combination of RB and CLAY. SBD with low values, also ISM, SCP, SNP, CLAY, and SAND with high values could dramatically interact with root properties, thus influencing the changes in PFI jointly (Fig. 10).

4. Discussion

4.1. Mechanism of soil physical properties driving preferential flow

4.1.1. Soil bulk density

Consistent with previous studies of Paradelo et al. (2016), our results indicated that high soil bulk density promotes the occurrence of preferential flow (Figs. 5a, b, c). The mechanism was that high soil bulk density below a threshold could reduce soil capillary porosity, resulting in dominant water flow passing larger pores and promoting the occurrence of preferential flow (Koestel et al., 2013). Another explanation was that high soil bulk density below a threshold might lead to the continuity of non-capillary pores due to the reduction of soil capillary pores function, which is conducive to the occurrence of preferential flow (Mossadeghi-Björklund et al., 2015). However, our results were inconsistent with the findings of Wei et al. (2024), Kang et al. (2022), and Fang et al. (2024), who reported that high soil bulk density could reduce the strength of preferential flow. This is because high soil bulk density above a threshold might decrease the number of macropores, limiting the occurrence of preferential flow (Mossadeghi-Björklund et al., 2015). Therefore, the effects of soil bulk density on preferential flow are contradictory and might be different at various sites or landscape scales.

4.1.2. Initial soil moisture

In agreement with the studies of Kang et al. (2022), Tang et al. (2020), and Demand et al. (2019), the results revealed a negative correlation between initial soil moisture and preferential flow (Figs. 5a, b, c). The possible explanation might be that water could preferentially transport through limited macropores, bypassing the soil matrix in soils with smaller initial soil moisture (Kobayashi and Shimizu, 2007). However, our results also showed that initial soil moisture might positively affect preferential flow (Fig. 9d), which is similar to the findings of Luo et al. (2022), Kung et al. (2006) and Jaynes et al. (2001). The possible explanation was that higher initial soil moisture in soils with low sand content could increase soil hydrophobicity, which was conducive to the occurrence of preferential flow (Kang et al., 2022). Therefore, the interaction between initial soil moisture and preferential flow was complex and contradictory, that is, initial soil moisture might interact with soil texture to affect the occurrence of preferential flow comprehensively.

4.1.3. Soil porosity

In agreement with the results of Zhang et al. (2024) and Qiu et al. (2022), our results indicated that SNP was conducive to the occurrence of preferential flow at three forest stands, while SCP reduced

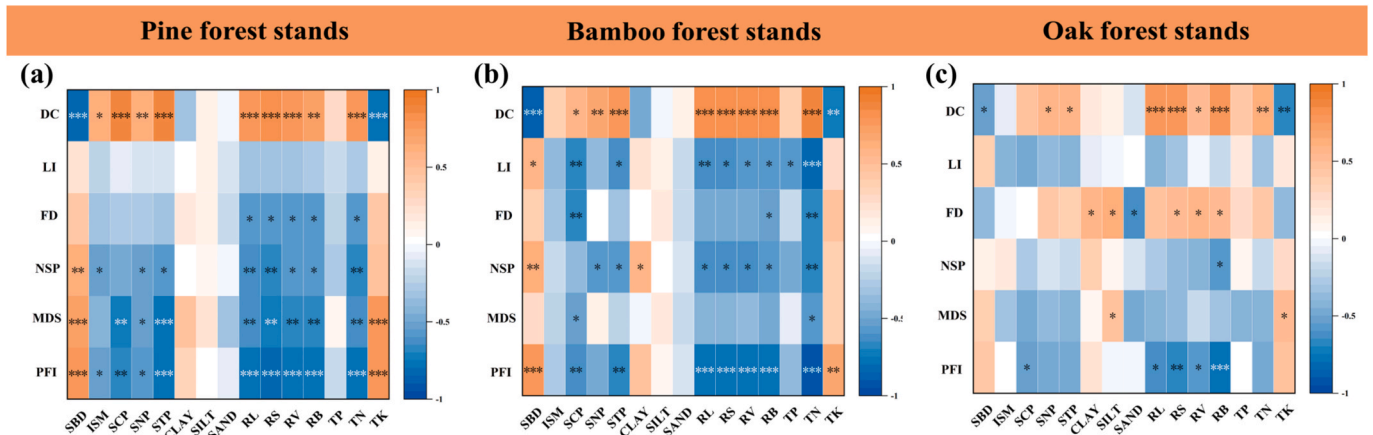


Fig. 5. (a), (b), (c): Pearson correlation analysis between soil physical properties, root properties, and soil nutrient properties and *PFI* at different forest stands.

preferential flow intensity (Figs. 9b, d, f). The possible explanation might be that well connected macropores are important paths for preferential flow and could enhance soil infiltration capacity, while *SCP* could drive slow water movement through the soil matrix by slow convective flux (Tang et al., 2024; Kang et al., 2022; Wang et al., 2019). Many studies indicated that high *STP* could enhance preferential flow intensity, which consistent with our results at the pine and oak forest stands (Qiu et al., 2022; Chen et al., 2021) (Figs. 9b, f). However, the results at bamboo forest stands were the contrary. The possible mechanism was that high soil porosity caused by abundant roots could enhance the formation of connected pore spaces, resulting in uniform soil infiltration (Yang et al., 2023; Kang et al., 2022) (Fig. 9d). Another reason might be that soil water holding capacity increases with increasing soil porosity, implying weaker preferential flow intensity (Zhuo et al., 2010). Both the positive and negative impacts of soil porosity in our study area might be attributed to a threshold controlling preferential flow intensity. Specifically, soil porosity might promote preferential flow intensity when it is below a threshold. Conversely, once soil porosity exceeded the threshold, soil porosity might reduce preferential flow intensity. In our study area, the soil porosity at the bamboo forest stand should exceed the threshold, promoting water infiltration into the soil matrix, which might reduce the occurrence of preferential flow (Yang et al., 2023).

4.1.4. Soil texture

Our results showed that high clay content was conducive to the occurrence of preferential flow, which is consistent with the studies of Grant et al. (2019a), Grant et al. (2019b), and Glæsner et al. (2011). A possible explanation might be that water could flow rapidly to the subsoil through shrinkage cracks when the clay soils become dry (Dekker and Ritsema, 1996). Another reason might be that cracks in dry clay soils could be connected to other preferential flow paths (i.e., root channels, and worm burrows), increasing the connectivity of pores and the strength of preferential flow (Kan et al., 2020; Grant et al., 2019a; Ernst et al., 2009). However, Grant et al. (2019a) and Zhang et al. (2024) found that clay content positively affected dye-stained coverage, which indicates that clay content is not conducive to the occurrence of preferential flow. The possible mechanism might be that clay content could reduce soil aggregate stability and the development of macropores (Zhang et al., 2024; Hudek et al., 2017). Soil aggregate stability and macropores are conducive to the occurrence of preferential flow (Kang et al., 2022; Wuest, 2009). Inconsistent with the results of Ding et al. (2023) and Liu et al. (2021), our results indicated that there was no significant correlation between sand content and preferential flow. Therefore, the relationships between soil texture and preferential flow are complex and contradictory.

4.2. Mechanism of root properties driving preferential flow

Consistent with Fang et al. (2024), high *RL* could reduce the occurrence of preferential flow at the bamboo forest stand (Figs. 9c, d). The possible mechanism might be that the fibrous root systems at the bamboo forest stand are distributed in a radial pattern, which results in the larger number of fibrous root systems than the other two stands. Fibrous root systems enlargement at the bamboo forest stand could promote the formation of excessive and overly large pores and disrupt the structure of the soils, making the connectivity of soil pores more uniform, which might result in uniform water infiltration (Yang et al., 2023; Patra et al., 2022). At the pine forest stand, high *RV* was the most important root morphological trait reducing the occurrence of preferential flow, which supports the results of Liu et al. (2020). The root systems of the pine forest stand are characterized by a high proportion of coarse root systems and deep vertical distribution within the soils (Yan et al., 2013). High *RV* could improve pores connectivity in the upper soil layers, promoting uniform water infiltration (Liu et al., 2020). Similar with the studies of Fang et al. (2024) and Zhang et al. (2015), high *RB* at the oak forest stand was not conducive to the occurrence of preferential flow. The possible explanation was that high *RB* could rearrange soil particle and reduce connectivity between pores (Wang et al., 2025). Therefore, based on the different mechanism for the three kinds of vegetation types, we inferred that root morphological traits controlling the occurrence of preferential flow might be dramatically various. Also, different root morphological traits had either positive or negative influences on preferential flow (Fig. 9), which indicated the complex relationship between root systems and preferential flow.

Besides, the joint effects of soil- and root properties on preferential flow should not be ignored (Fig. 10) (Zhu et al., 2025). Root systems could improve soil structure and reduce soil bulk density, affecting the occurrence of preferential flow (Wei et al., 2024; Jiang et al., 2020). Soils with low soil bulk density are more likely beneficial to root enlargement, promoting the effective connections between the soil pore networks (de Moraes et al., 2020). In this study, high initial soil moisture could be conducive to root growth, which increases the even distribution of the root-soil pores, negatively affecting the occurrence of preferential flow (Chun et al., 2021; Zhang et al., 2016). Soil porosity could be improved with excessive root development, which results in uniform water infiltration, reducing the occurrence of preferential flow (Dimattia et al., 2025; Zhang et al., 2021a). High clay content could enhance the soil's ability to obtain nutrients and further promote to develop stronger root systems, disrupting the soil structure and increasing preferential flow intensity (Alhaj Hamoud et al., 2019). In summary, the interaction between root systems and soil properties was complex, which results in different mechanism driving preferential flow in the three kinds of vegetation types.

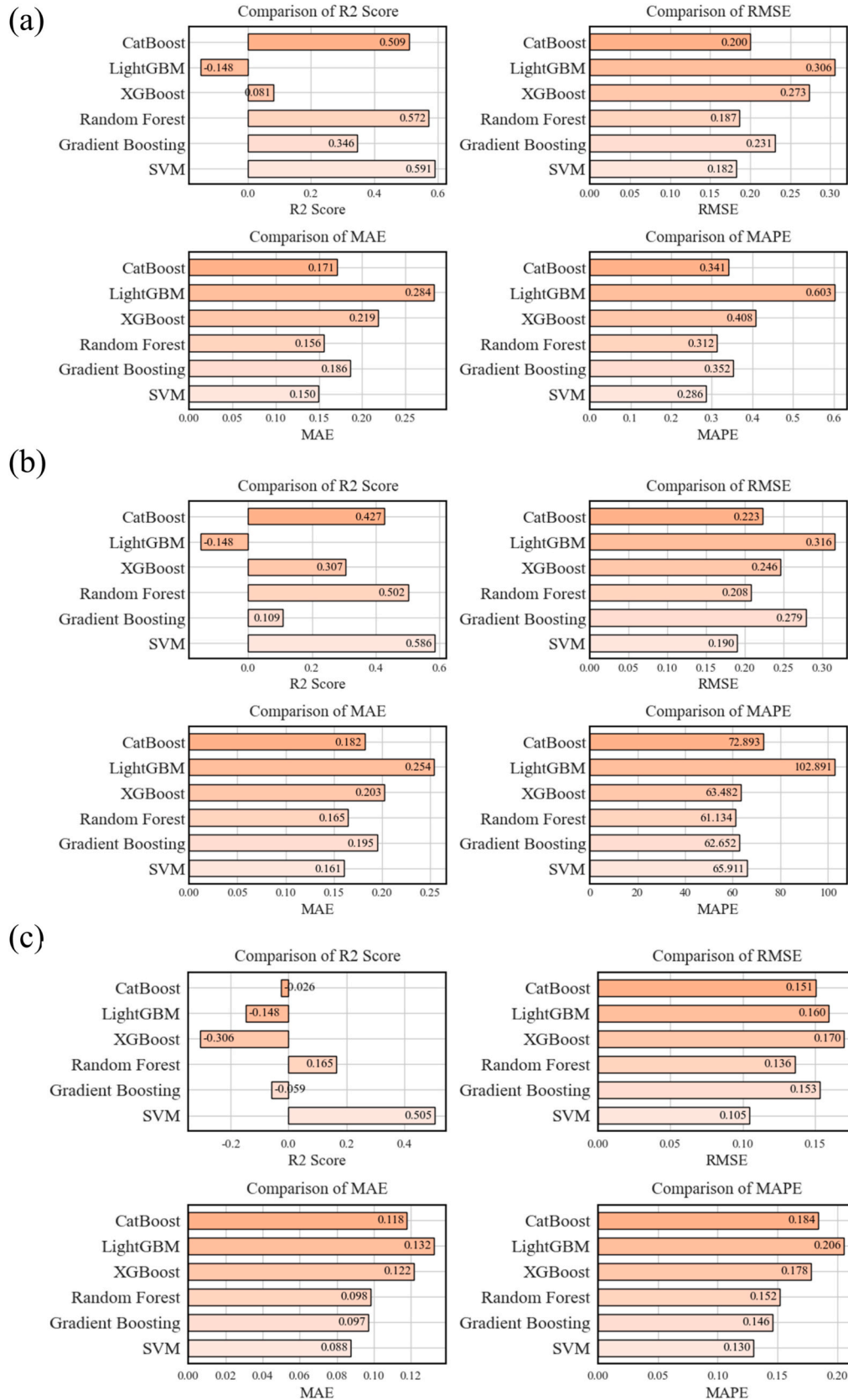


Fig. 6. Comparison of performance of different models at (a): pine forest stand, (b): bamboo forest stand, and (c): oak forest stand.

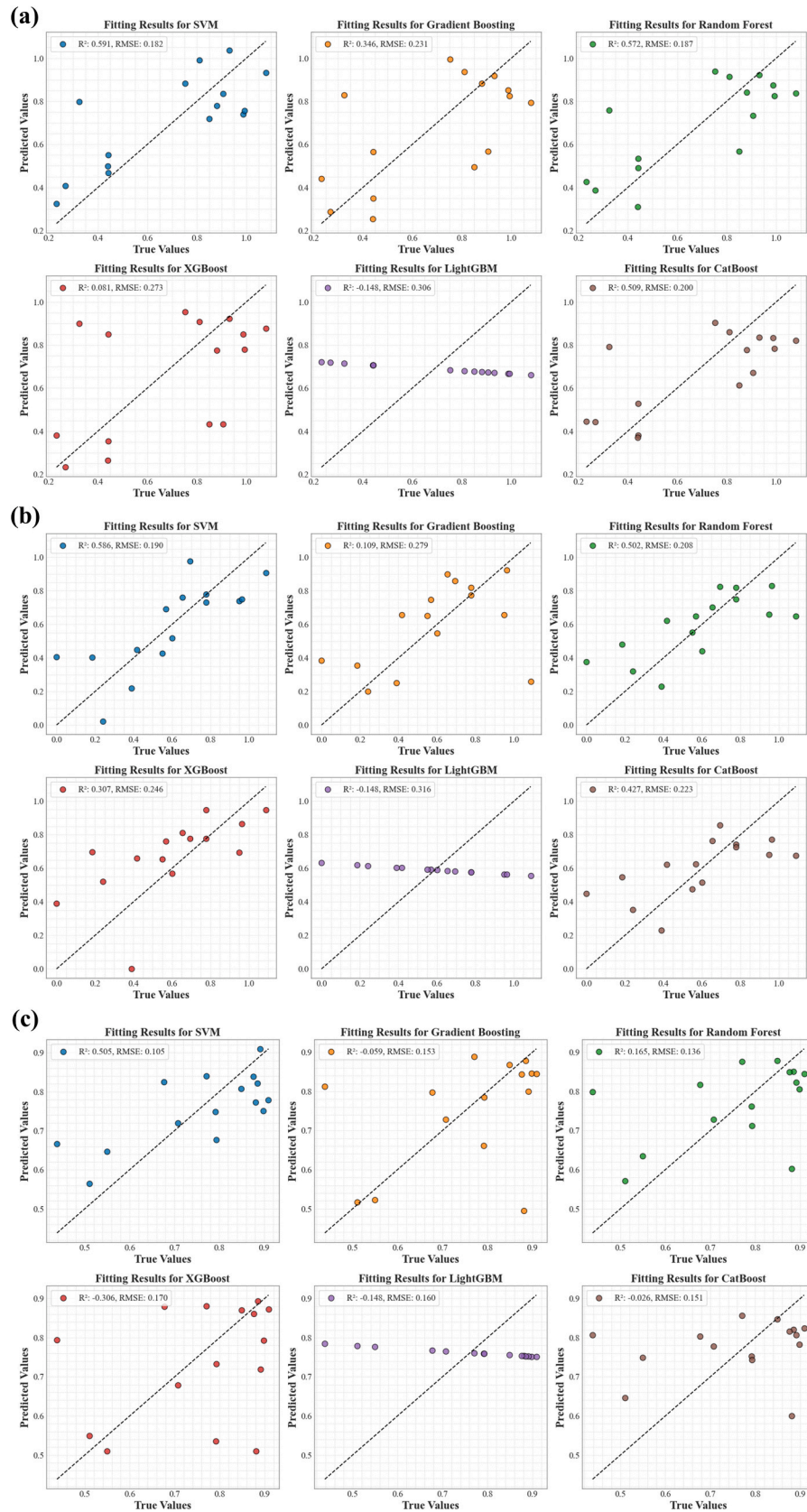


Fig. 7. Comparison of fitting results of different models at (a): pine forest stand, (b): bamboo forest stand, and (c): oak forest stand (the predicted and true values are *PFI*).

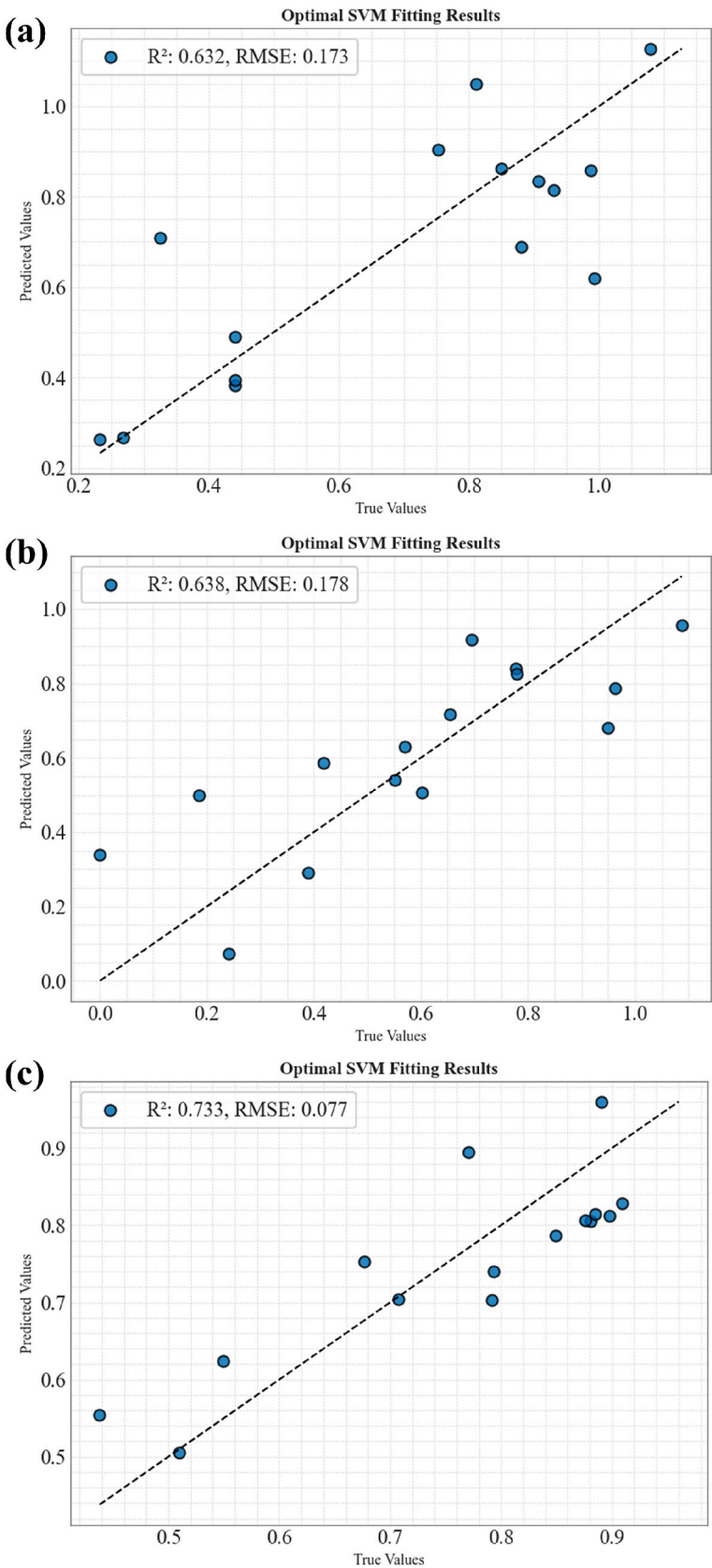


Fig. 8. The results of further optimizing the SVM with PSO algorithm at (a): pine forest stand, (b): bamboo forest stand, and (c): oak forest stand.

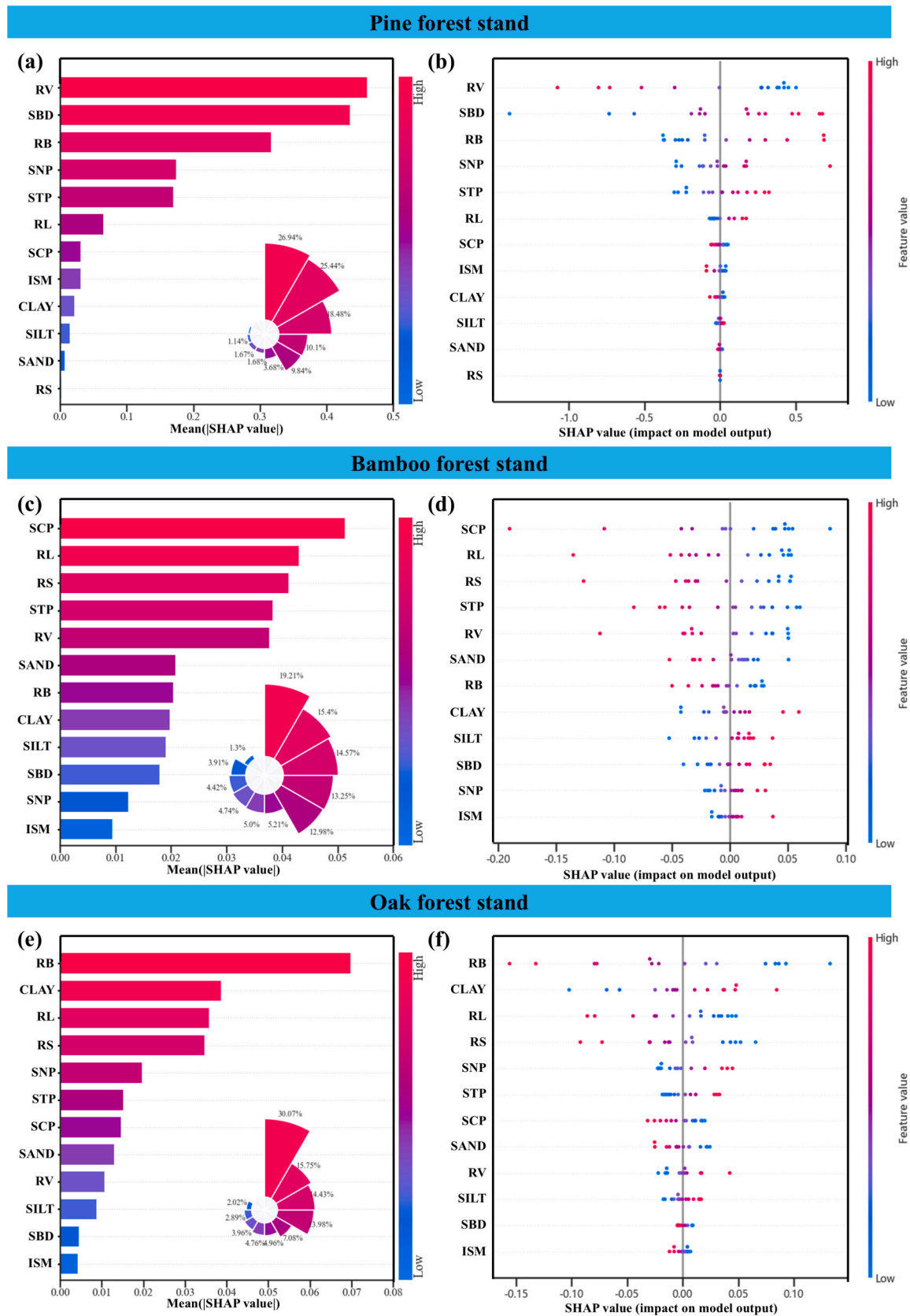


Fig. 9. (a), (c), (e): Global SHAP importance plots: the horizontal bars show the mean absolute SHAP value for each variable, ranking the average importance of soil- and root properties on the predicted *PFI*. (b), (d), (f): SHAP summary plots: each point represents an individual observation, which could be plotted as the variable's actual values (color scale) corresponding to SHAP values (x-axis), revealing both the direction and magnitude of the contribution of each variable to the model output at the three forest stands.

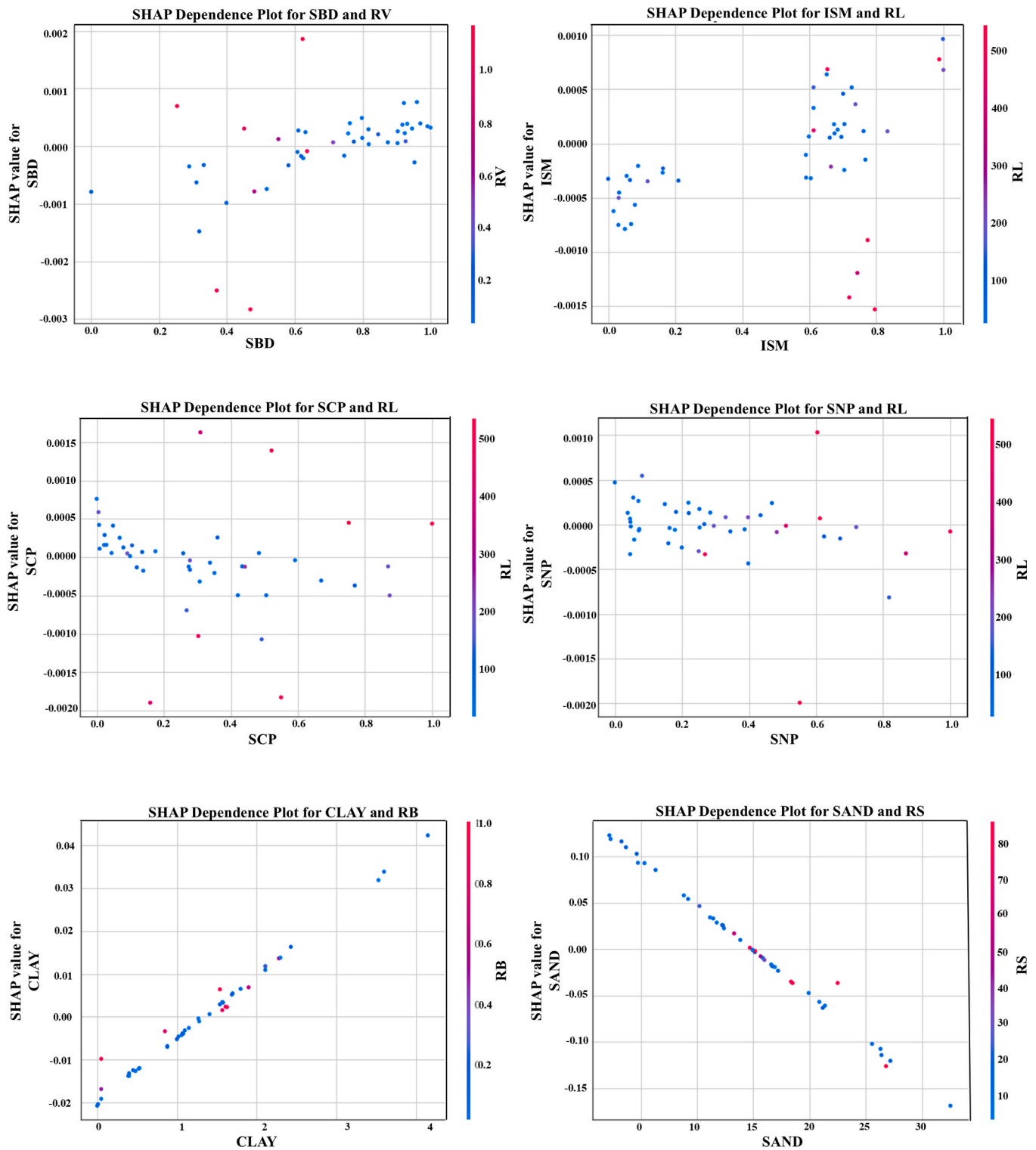


Fig. 10. SHAP dependence plot for different variables at the three forest stands.

4.3. The role of different vegetation types in preferential flow

Consistent with the studies of Zhang et al. (2023), our results showed that preferential flow intensity at the broadleaved forest stands (oak) was higher than that at the coniferous forest stands (pine) and bamboo forest stands in the upper soil layers. This is because that the decomposition rate of broadleaved residues is faster than others, improving soil properties in the upper soil layers, thereby promoting the occurrence of preferential flow (Zhu et al., 2021; Chae et al., 2019; Piaszczyk et al.,

2019). Meanwhile, the differences in the canopy interception intensity of different vegetation types could also lead to the variations in preferential flow intensity. Studies showed that the canopy interception of the bamboo forest stand was lower than the broadleaved and coniferous forest stands (Komatsu et al., 2010). Therefore, the soils could be more likely to reach soil moisture saturation because of the lower canopy interception of the bamboo forest stand. Studies showed that high soil moisture could reduce preferential flow intensity and increase the uniform soil infiltration (Kang et al., 2023). In our study area, there was a

sudden decrease in the soil clay content at the delimitation (20 cm) of the upper soil layers versus lower layers at the pine and oak forest stands. Decreasing soil clay content reduces the stability of soil aggregates, which tends to make soil pore structure could be easily destroyed, reducing preferential flow intensity (Amezketta et al., 2004). The relationships between soil properties, root properties, and preferential flow were consistent both in the upper soil layers (0–20 cm) and the lower soil layers (20–50 cm) (Fig. S1). Therefore, we inferred that the sudden decrease in the soil clay content might be the main factors contributing to the differences in preferential flow intensity with increasing soil depth between the pine and oak forest stands (Fig. 4), which supports the studies of Zhang et al. (2024).

4.4. The choice of best machine learning model and SHAP explanation

In agreement with the studies of Ahmad et al. (2010) and Yu et al. (2012), our results showed that SVM model performed the best compared with the other five models (Figs. 6, 7, 8). In general, the variables influencing the changes in preferential flow were numerous and complex, which jointly could constitute a high-dimensional data set. Also, the number of samples obtained from field experiments is limited. SVM model could effectively deal with high-dimensional dataset, which mitigates the increased computational complexity and overfitting (Chauhan et al., 2018; Muriira et al., 2018; Erfani et al., 2016). Meanwhile, compared with the other models, SVM model could fully utilize the limited information of each sample in small datasets (Erfani et al., 2016). SHAP could evaluate the contribution of different variables to the model output results and the direction of variables. SHAP could also quantify the complex and the non-linear joint effects of variables on model output results (Parsa et al., 2019). Consistent with the results of Wang et al. (2022) and Ding et al. (2025), our study also concluded that SHAP is available to identify the effects of driving factors in the studies of hydrology. In summary, the model developed in our study could predict the occurrence of preferential flow with high accuracy, and also offer clear insights into soil- and root properties driving preferential flow in other study areas.

4.5. Limitations and future considerations

Although the SVM model performance has been greatly improved by optimizing with PSO (Fig. 8), PSO algorithm might obtain an optimal result with the local best parameters (Yao et al., 2024). In future, we should fully consider hybrid optimization algorithms (i.e., genetic algorithms, differential evolution algorithms, annealing algorithm, etc.) in the application to the SVM model to improve the effectiveness of parameter optimization and obtain the global best parameters (Kaplan et al., 2024; Alhijawi and Awajan, 2023; Bilal et al., 2020). Different forested landscapes have distinct variations in topography, vegetation cover, and soil properties, which are crucial factors influencing preferential flow (Qiu et al., 2022; Garrido and Helmhart, 2011; Sidle et al., 2001). Cai et al. (2024) concluded that preferential flow intensity increased with increasing forest age due to enhanced soil structure and increased root biomass, which promotes connectivity of soil macropore networks. In future, in order to further explore the driving mechanism of preferential flow, we should consider more forested landscapes and the importance of forest age. Seasonal variations also impact the occurrence of preferential flow. Our field experiments were conducted in autumn, which limits our understanding of the mechanisms of seasonal variations effects. For instance, Demand et al. (2019) observed that preferential flow intensity is low in spring and high during summer and early autumn. Tăumer et al. (2006) found that preferential flow intensity peaks in summer and early autumn. Therefore, further studies across multiple seasons should be fully considered.

Consistent with the studies of Yan et al. (2023), our results showed that *PFI* was negatively correlated with *TN* and *TP*, while it had positive effects on *TK* (Fig. 5 a, d, g). The possible reason might be that soluble

nitrogen and phosphorus are easily transported through preferential flow paths, while potassium as a non-soluble substance is largely retained along the path walls (Duan et al., 2025; Wang et al., 2018). Slope position, vegetation types, and precipitation significantly influence these processes (Duan et al., 2025; Yan et al., 2023; McGrath et al., 2010). For example, middle slopes with dense herbaceous-shrub communities exhibit stronger vertical transport, leading to greater nitrogen and phosphorus loss but higher potassium retention compared to grassy upper or lower slopes (Li et al., 2024). Furthermore, preferential flow during rainfall events could also increase nutrient loss risk, particularly in surface soil layers on upper slopes (Duan et al., 2025). Therefore, it is necessary to consider topographical characteristics to conduct forestry management so that managers could better mitigate the negative effects of preferential flow on soil nutrients losses. We also suggest that forest managers could conduct thinning, selective logging, and shallow-rooted grass planting to enhance lateral water flow and increase soil nutrient residence time, thereby promoting nutrient sustainability and forest ecosystem health (Shah et al., 2022; Jotisankasa and Sirirattanachai, 2017).

5. Conclusions

This study clarified the mechanisms of soil- and root properties driving preferential flow at the pine, bamboo, and oak forest stands. The main findings revealed that preferential flow intensity increased with increasing soil depth. The preferential flow intensity in the topsoils was the highest at the pine forest stand, while it was the highest at the oak forest stand in the subsoils. The SVM model showed superior performance in explaining the complex relationships between soil properties, root properties, and preferential flow. Crucially, the primary driving factors of preferential flow were various at different forest stands: *RV* showed the highest contributions to the occurrence of preferential flow at the pine forest stand, while *SCP* and *RB* at the bamboo and oak forest stands, respectively. We also further identified that the interactions between soil properties and root systems might also control the occurrence of preferential flow. This study provided theoretical insights into the manners of preferential flow governing soil nutrients at forest stands and offered scientific evidence for soil water management in forest ecosystems.

6. Availability of data

Data are available on request from the authors.

CRedit authorship contribution statement

Yiyan Liu: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Le You:** Software, Methodology, Investigation. **Wenqi Zhang:** Validation, Resources, Data curation. **Zhiying Tang:** Supervision, Resources, Data curation. **Di Wang:** Supervision, Resources, Data curation. **Yinghu Zhang:** Writing – review & editing, Validation, Supervision, Investigation, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.catena.2025.109544>.

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